Highlights

Domain-anchored DINOv2 representation learning for oriented poultry-house detection from aerial imagery: a multi-seed development study

Anonymous Authors

- A frozen DINOv2 teacher adapted the YOLO11l-OBB backbone using unlabeled regional aerial imagery.
- The proposed method improved the three-seed mean by 0.00212 on validation and 0.00437 on the observed test.
- The all-seed validation criterion was not met; uncertainty is reported rather than hidden.
- Pseudo-label transfer from Alabama to Mississippi was label-starved and class-collapsed.
- Geographic, farm, or year-disjoint evaluation is required before a deployment claim.

Domain-anchored DINOv2 representation learning for oriented poultry-house detection from aerial imagery: a multi-seed development study

Anonymous Authors[a]

[a] *Affiliation withheld for blinded review, United States*

Abstract

Context: Reliable aerial inventories of poultry houses can support biosecurity, environmental assessment, and agricultural planning. Their elongated, rotated footprints make oriented object detection appropriate, yet annotated imagery is limited and spatially heterogeneous. Unlabeled regional imagery is abundant, but it is not clear whether it can improve an already pretrained detector without destabilizing useful detection features.

Objective: We evaluated whether domain-anchored DINOv2 feature distillation from unlabeled Mississippi National Agriculture Imagery Program (NAIP) tiles improves YOLO11l oriented bounding-box detection of poultry houses in Alabama.

Methods: A frozen DINOv2 ViT-B/14 teacher guided adaptation of the YOLO11l-OBB backbone on a fixed 250,000-image Mississippi RGB corpus. Multi-scale cosine alignment at YOLO feature levels P3, P4, and P5 was constrained by a relative parameter-anchor penalty to the original detector initialization. Disposable adaptation projectors were removed before supervised fine-tuning. The proposed initialization and a matched Ultralytics YOLO11l-OBB baseline were trained for 100 epochs at 1024 pixels with seeds 7, 17, and 42 on deduplicated Alabama development data. Checkpoints were selected using validation only, frozen by SHA-256 hash, and then evaluated once on the held test files.

Results: Anchored adaptation produced a favorable but inconclusive validation mean: 0.86489 ± 0.00528 versus 0.86278 ± 0.00133 for the matched baseline. On the frozen observed test, the corresponding means were 0.80662 ± 0.00257 and 0.80224 ± 0.00336. Seed 42 improved both validation (0.86511 versus 0.86328) and the observed test (0.80953 versus 0.80023). However, the

paired 95% intervals included zero for validation and test, and the validation rule requiring improvement in every seed was not met. A separate pseudo-labeling study retained labels on only 0.195–0.342% of sampled Mississippi images and collapsed to one class, explaining why pseudo-label branches were not combined with the proposed method.

Conclusions: Domain-anchored DINOv2 adaptation is a technically coherent and promising route for poultry-house OBB detection, but the present evidence does not establish a robust superiority claim. The current test files are byte-disjoint, yet every test source group recurs in development and near-duplicate structure is present. A geographic, farm, or year-disjoint test is therefore required before claiming generalization or deployment readiness.

Keywords: Poultry-house mapping, Self-supervised learning, Oriented object detection, Aerial imagery, DINOv2, Domain adaptation

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1. Introduction

Industrial poultry production is spatially concentrated, operationally consequential, and difficult to inventory consistently from ground observation alone. Aerial image analysis offers a practical route to mapping poultry facilities at scale, as demonstrated by prior work that combined deep learning and high-resolution imagery to identify industrial poultry operations (Robinson et al., 2021). For this application, a detector must identify buildings whose visual signature is dominated by long rooflines, repeated farm layouts, and arbitrary orientation. Oriented bounding boxes (OBBs) are therefore preferable to axis-aligned boxes when the goal is to preserve the footprint geometry needed for downstream mapping.

The central methodological problem is not whether large image models are useful in the abstract. It is whether unlabeled imagery from the region in which a system will operate can improve a strong detector initialization without erasing the geometric priors already encoded by that detector. NAIP provides a large archive of agricultural aerial imagery (U.S. Geological Survey, 2026); it is attractive as an unlabeled source, but regional imagery differs from labeled target imagery in acquisition date, tile construction, land cover, roof appearance, and class frequency. Straightforward pseudo-labeling can amplify that mismatch when a source-trained detector rarely recognizes target objects with high confidence. Semi-supervised OBB methods have made substantial progress when labeled and unlabeled images share a task distribution (Hua

et al., 2023), but the same assumption is fragile under regional shift.

We study a narrower and more defensible alternative: *domain-anchored DINOv2 adaptation*. A frozen DINOv2 vision transformer provides dense self-supervised features, while a YOLO11l-OBB backbone is adapted on unlabeled Mississippi imagery under an explicit penalty that keeps it near its original detector initialization. The adapted backbone is then placed back into the original detector and fine-tuned with Alabama labels. This is not DINOv2 pretraining from scratch, and it is not a claim that a generic foundation model replaces a detector. It is a constrained transfer procedure that asks whether an existing aerial detector can absorb region-specific visual information while retaining its detection competence.

The paper makes four contributions.

1. It specifies and evaluates a domain-anchored DINOv2 feature-distillation protocol for transferring unlabeled regional aerial imagery into an OBB detector.
2. It reports the complete three-seed comparison against an otherwise matched YOLO11l-OBB baseline, including the seed that did not improve and paired uncertainty intervals.
3. It treats pseudo-labeling as a diagnostic branch rather than an automatic ingredient: a retention audit shows that the Alabama-to-Mississippi pseudo-label signal was too sparse and class-imbalanced to justify stacking it with the anchored method.
4. It documents a strict separation between model selection and test evaluation, while explicitly identifying why the currently available test split is insufficient for a claim of geographic generalization.

The fourth point is deliberate. Agricultural computer vision is especially vulnerable to inflated accuracy when nearby tiles, repeated acquisition areas, or augmented views appear across splits. Our goal is to make the present evidence useful without overstating it. The proposed method is the main scientific contribution because it supplies a coherent regional representation-learning story; it is not presented as a definitive performance winner until it is evaluated on a genuinely independent geographic, farm, or year-disjoint holdout.

2. Related work

2.1. Aerial detection of agricultural infrastructure

Poultry houses are a practical target for remote sensing because their shape and clustering encode facility-level agricultural activity. The operational promise of mapping has to be balanced against the difficulty of transferring a detector across imagery sources and locations. High-resolution aerial benchmarks such as DOTA established the importance of orientation-aware localization for objects that occur at arbitrary angles (Xia et al., 2018). Poultry buildings sharpen the problem: they are often visually simple in isolation but occur amid confounders such as warehouses, greenhouses, roads, shadow, vegetation, and roof-state variation.

2.2. Self-supervision and teacher-guided transfer

DINOv2 showed that self-supervised visual features can transfer broadly across vision tasks (Oquab et al., 2023). However, a strong feature teacher does not by itself solve the interface problem between a transformer embedding space and a detector backbone. In this study, the teacher is used as a fixed source of dense patch features. The student remains the actual YOLO backbone, and its output is constrained to remain close to the original detector parameters. This design targets a practical failure mode in domain adaptation: unconstrained unlabeled adaptation can overwrite useful detector features before any target labels are seen.

2.3. Why pseudo-labeling is a comparison, not a default

Pseudo-labeling has become a standard route to exploit unlabeled images, including for oriented detection (Hua et al., 2023). Its quality depends on enough confident predictions to create a representative learning signal. Under a marked shift in scale, appearance, or class prevalence, filtering can instead produce a tiny and biased label set. For that reason, pseudo-labeling was evaluated as a separate branch with matched continuation controls and a retention audit. It was not combined with anchored adaptation after the audit showed that the required signal was absent.

3. Materials and experimental boundary

3.1. Labeled Alabama data

The supervised task contains two OBB classes: *broiler breeder* and *broiler house*. To prevent the validation comparison from benefiting from exact

duplicate imagery, the development manifests were deduplicated by SHA-256. The resulting development set contains 4,139 training images and 442 validation images, reduced from 4,359 and 465 raw images, respectively; no exact train-validation overlap remained. The labeled development imagery is from Alabama and is used only for detector training and model selection.

The available held test files contain 702 Alabama images. They were intentionally excluded from the training configuration and were evaluated only after the validation checkpoints were frozen. That separation protects model selection from direct test tuning, but it does not make the test geographically independent. The raw split audit found zero exact SHA-256 matches between training and test files, yet all 56 test source groups recur in development: 52 in raw training and the remaining four in validation. A perceptual-hash audit also flagged 134 of 702 test images as near to a training image at Hamming distance at most 8, with a minimum distance of 2. We therefore call these measurements the *observed AL test* or a tile-level held test throughout the paper. They must not be interpreted as cross-farm, cross-year, or cross-region generalization.

3.2. Unlabeled Mississippi imagery

The unlabeled source was a fixed manifest of 250,000 RGB tiles sampled from Mississippi NAIP-derived imagery. The manifest was seeded before training and split by source group into 237,488 training images from 3,075 groups and 12,512 internal validation images from 161 groups, with no group overlap. The source corpus contains no Alabama labels and no held-test files. The pretraining stage used 448-pixel random crops; this is a representation-adaptation stage, not a source-label training stage.

Table 1: Data roles and separation audit. Development data are used for training and validation-only checkpoint selection. The observed test is held out from selection but is not source-group independent.

Corpus	Role	Content and scale	Separation condition
Mississippi NAIP-derived corpus	Unlabeled representation adaptation	250,000 RGB tiles; 237,488 train and 12,512 internal validation images	3,075 training and 161 validation source groups; no group overlap
Alabama development corpus	Detector training and checkpoint selection	4,139 train and 442 validation images; two OBB classes	Exact SHA-256 duplicates removed across development manifests
Observed Alabama test	Frozen checkpoint evaluation only	702 images; no test image used in selection	Byte-disjoint, but all 56 source groups recur in development; 134/702 images are near to a training image at pHash distance ≤ 8

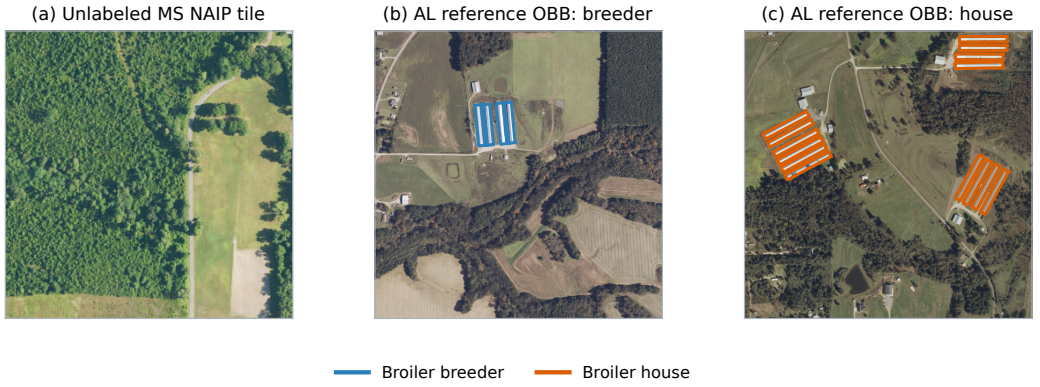


Figure 1: Data roles and image appearance. (a) A representative unlabeled Mississippi NAIP-derived tile used only for representation adaptation. (b–c) Alabama development examples with reference OBBs for the two poultry-house classes. The colored polygons are original annotations; no held-test image is shown.

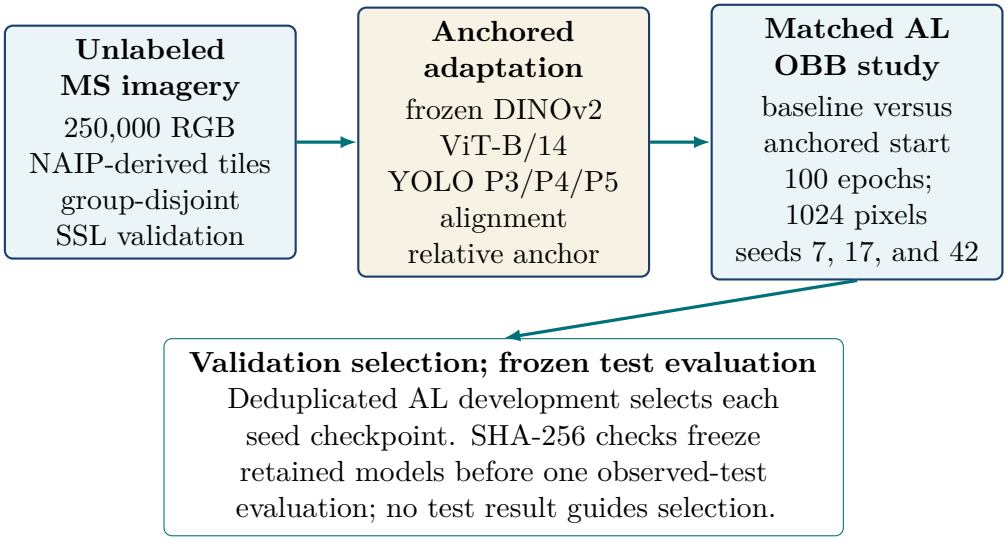


Figure 2: Study design. The proposed method transfers a constrained, teacher-guided backbone adaptation from unlabeled Mississippi imagery into a matched Alabama OBB experiment. The test separation is procedural, but the source-group and near-duplicate audit means the observed test is not a geographic independence test.

4. Methods

4.1. Domain-anchored DINOv2 feature distillation

The adaptation stage used no Alabama labels. Let $\boldsymbol{\theta}^{(0)} = \{\boldsymbol{\theta}_j^{(0)} : j \in \mathcal{J}\}$ denote the parameter tensors of the pretrained YOLO11l-OBB backbone shared by the two supervised arms, and let $\boldsymbol{\theta}$ denote their values during adaptation. We optimized only YOLO modules 0–9, which form the detector backbone and contain 11,377,984 trainable parameters. The teacher was the frozen `vit_base_patch14_dinov2.lvd142m` model. After normalization with its pretrained image statistics and removal of its prefix token, the teacher mapped each 448×448 RGB crop to a 32×32 grid of 768-dimensional patch representations. No teacher parameter was updated.

The adaptation objective transfers dense teacher structure to three student feature scales while preserving each scale's native spatial resolution. Let $\mathcal{S} = \{\mathrm{P3}, \mathrm{P4}, \mathrm{P5}\}$. For a mini-batch $\mathbf{X}$, the student feature map at scale s is $\mathbf{F}_s(\mathbf{X}; \boldsymbol{\theta})$. A learned 1×1 projector $\Pi_s(\,\cdot\,; \boldsymbol{\phi}_s)$ maps that feature map to 768-dimensional tokens, and $\mathcal{R}_s$ bilinearly resizes the teacher token grid to the corresponding student resolution with `align_corners=False`. At an aligned batch-spatial position q, let $\mathbf{z}_{s,q} = [\Pi_s(\mathbf{F}_s(\mathbf{X}; \boldsymbol{\theta}); \boldsymbol{\phi}_s)]_q$ and $\mathbf{t}_{s,q} = [\mathcal{R}_s\{\mathcal{T}(\mathbf{X})\}]_q$ denote the projected student and frozen teacher vectors, respectively. We then minimized the scale-weighted mean cosine dissimilarity,

$$\begin{aligned} \mathcal{L}_{\mathrm{feat}}(\boldsymbol{\theta}, \boldsymbol{\phi}) &= \frac{\sum_{s \in \mathcal{S}} w_s \, (1 - \bar{c}_s)}{\sum_{s \in \mathcal{S}} w_s}, \\ \bar{c}_s &= \frac{1}{|\Omega_s|} \sum_{q \in \Omega_s} \frac{\langle \mathbf{z}_{s,q}, \mathbf{t}_{s,q} \rangle}{\max(\|\mathbf{z}_{s,q}\|_2, \epsilon_{\cos}) \max(\|\mathbf{t}_{s,q}\|_2, \epsilon_{\cos})}, \end{aligned} \tag{1}$$

where $\boldsymbol{\phi} = \{\boldsymbol{\phi}_s : s \in \mathcal{S}\}$ is the collection of projector parameters, w_s is the scale weight, $\bar{c}_s$ is the mean stabilized cosine similarity at scale s, Ω_s is the set of batch-spatial positions at scale s, $\mathcal{T}$ is the frozen DINOv2 teacher, and $\epsilon_{\cos} = 10^{-8}$ is the numerical floor used in cosine similarity. The scale weights were $w_{\mathrm{P3}} = 0.5$, $w_{\mathrm{P4}} = 1.0$, and $w_{\mathrm{P5}} = 1.0$.

A feature-only objective supplies no explicit constraint on how far the detector backbone may move from its pretrained detection representation. We therefore penalized relative parameter drift, giving equal weight to each named backbone parameter tensor rather than to each scalar parameter,

$$\mathcal{L}_{\mathrm{anch}}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \frac{n_j^{-1} \left\| \boldsymbol{\theta}_j - \boldsymbol{\theta}_j^{(0)} \right\|_{\mathrm{F}}^2}{\max\left(n_j^{-1} \left\| \boldsymbol{\theta}_j^{(0)} \right\|_{\mathrm{F}}^2, \epsilon_{\mathrm{anch}} \right)}, \tag{2}$$

where $\mathcal{J}$ indexes the named trainable backbone parameter tensors, $\boldsymbol{\theta}_j$ and $\boldsymbol{\theta}_j^{(0)}$ are the current and initial versions of tensor j, n_j is its number of scalar entries, and $\epsilon_{\text{anch}} = 10^{-6}$ prevents division by a negligible initial tensor energy. Both the numerator and reference scale were evaluated in FP32.

The complete adaptation objective combines feature agreement with this relative anchor,

$$\mathcal{L}_{\text{SSL}}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \mathcal{L}_{\text{feat}}(\boldsymbol{\theta}, \boldsymbol{\phi}) + \lambda_{\text{anch}} \mathcal{L}_{\text{anch}}(\boldsymbol{\theta}), \qquad \lambda_{\text{anch}} = 0.05\,. \tag{3}$$

The adaptation used AdamW for 30 epochs, 1,000 training batches per epoch, at most 100 validation batches per epoch, two warm-up epochs, batch size 8, BF16 arithmetic, and gradient clipping at 1.0. Backbone and projector learning rates were 10^{-4} and 10^{-3}, respectively, with weight decay 0.05. The best artifact had an internal SSL validation loss of 0.15288. The projector parameters $\boldsymbol{\phi}$ were discarded after adaptation; only the verified backbone state was injected into the original detector before supervised fine-tuning.

4.2. Matched detector comparison

The baseline and anchored arms used the same initial YOLO11l-OBB checkpoint from the Ultralytics implementation (Jocher et al., 2024), labels, seeds, detector configuration, and augmentation schedule. Each was fine-tuned for 100 epochs at 1024 pixels with batch size 8, learning rate 0.001, weight decay 0.0005, and seeds 7, 17, and 42. The intended experimental difference was therefore limited to whether the backbone had first undergone the anchored unlabeled adaptation. The neck and detection head retained their original pretrained state in both arms before supervised training.

The primary metric was OBB mAP50-95, the mean average precision averaged across IoU thresholds from 0.50 to 0.95. For each seed, the checkpoint with the best validation mAP50-95 was retained. The protocol wrote configuration, dataset, artifact, and checkpoint SHA-256 hashes, and it rejected a test evaluation if a frozen checkpoint no longer matched its validation record.

4.3. Statistical reporting and selection rule

All reported means use the sample standard deviation across three paired seeds. We also report the paired mean difference and a two-sided 95% t interval; with $n = 3$, these intervals should be read as uncertainty descriptions rather than as precise inferential estimates. Selection was restricted to Alabama validation data. The protocol required a non-negative mean difference *and*

improvement in all three seeds before labeling the anchored method as eligible for a superiority claim. This all-seed rule was intentionally stricter than selecting the largest mean alone.

After validation, the complete set of seed-specific best checkpoints was frozen and evaluated on the observed AL test. No observed-test value was used to choose an architecture, a hyperparameter, or a checkpoint. The separation is useful, but it does not repair the source-group issue described above; the test is reported as a held-file observation, not an external validation.

4.4. Companion experiments

Three companion branches contextualize the main method. First, a matched 640-pixel follow-up tests whether any effect is resolution-specific. Second, a three-seed 640-pixel prediction ensemble evaluates a practical deployment candidate using concatenated raw predictions followed by class-aware rotated non-maximum suppression. This ensemble was assessed on validation only and is not treated as a test result. Third, CTQ and EMA-style pseudo-label branches were run separately with supervised continuation controls. Their purpose was diagnostic: to establish whether adaptation through pseudo labels had sufficient coverage and whether any apparent gain survived a frozen test evaluation.

5. Results

5.1. Optimization and checkpoint-selection dynamics

Figure 3 documents the two stages that precede the paired evaluation. The Mississippi adaptation objective fell from 0.38697 to 0.14033 on training batches and from 0.23658 to 0.15523 on the internal source-group validation stream. The retained backbone is the 23rd adaptation epoch, where the internal validation objective reached its minimum of 0.15288. The small late rise in validation loss is precisely why the artifact was selected by the internal validation minimum rather than by the final epoch.

During supervised fine-tuning, both methods reached a high validation plateau. The thin curves in Figure 3 retain the individual seed trajectories rather than implying that there is one deterministic learning curve per method. Anchored SSL did not establish faster convergence: it began lower and showed more between-seed variation. Its evidence is instead the final, seed-specific best-validation comparison reported below. The baseline run for seed 17

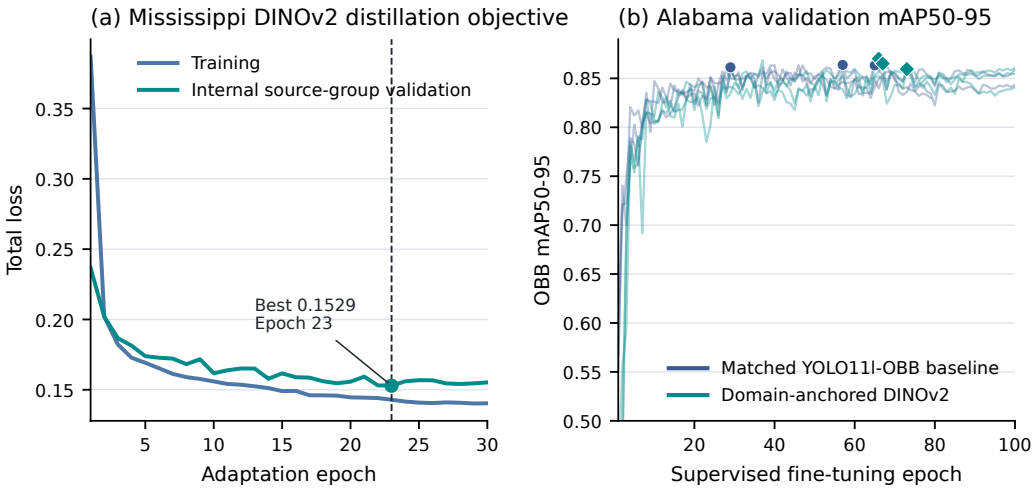


Figure 3: Optimization behavior of the anchored representation and the matched detector study. (a) Training and internal source-group validation loss during 30 epochs of Mississippi DINOv2 feature distillation. The retained backbone is the minimum validation-loss artifact at epoch 23. (b) Alabama validation mAP50-95 during 1024-pixel supervised fine-tuning. Thin paths are individual seeds; markers identify the checkpoint selected from each path. These panels document optimization and checkpoint selection, not a statistical superiority test.

stopped after 79 epochs under the configured patience rule; all other displayed runs reached the 100-epoch ceiling.

5.2. Primary 1024-pixel comparison

Table 2 gives the complete paired result. On validation, anchored DINOv2 adaptation raised the mean from 0.86278 to 0.86489, a mean paired difference of +0.00212. The result was not uniform: seed 7 improved by +0.00628, seed 42 improved by +0.00183, and seed 17 declined by −0.00176. Consequently, the validation 95% paired interval (−0.00789 to +0.01212) crossed zero and the all-seed rule was not satisfied.

The frozen observed-test comparison had the same directional mean pattern: 0.80662 for anchored SSL versus 0.80224 for the baseline, a paired mean difference of +0.00437. Two seeds favored anchored SSL, whereas seed 7 declined slightly. The observed-test 95% paired interval (−0.00915 to +0.01790) also crossed zero. Seed 42 is noteworthy as a descriptive paired result because it improved on both validation and test, but the table makes clear why it cannot be promoted to a confirmatory headline.

Table 2: Complete primary comparison at 1024 pixels. Values are OBB mAP50-95. Positive differences favor domain-anchored DINOv2 adaptation. The held test is reported as the observed AL test because it is not geographically or source-group independent.

Validation			
Seed	Baseline	Anchored	Difference
7	0.86378	0.87006	+0.00628
17	0.86127	0.85951	-0.00176
42	0.86328	0.86511	+0.00183
Mean ± SD	0.86278 ± 0.00133	0.86489 ± 0.00528	+0.00212
Paired 95% interval	[-0.00789, +0.01212]		
Observed AL test			
Seed	Baseline	Anchored	Difference
7	0.80612	0.80465	-0.00147
17	0.80038	0.80568	+0.00530
42	0.80023	0.80953	+0.00930
Mean ± SD	0.80224 ± 0.00336	0.80662 ± 0.00257	+0.00437
Paired 95% interval	[-0.00915, +0.01790]		

The internal protocol therefore recorded the baseline as the validation winner under its hard all-seed criterion. This is an important distinction: anchored DINOv2 is the main method studied here because it is the most coherent and auditable representation-learning contribution, not because the current three-seed outcome proves universal superiority. Its appropriate interpretation is a positive signal that warrants a stronger independent evaluation.

5.3. Qualitative OBB localization

Figure 5 makes the target task visible without replacing the aggregate analysis with selected images. The two Alabama validation scenes were fixed from annotation strata before inspection of predictions: one contains only broiler-breeder boxes and the other contains a dense broiler-house facility. For compactness, the figure uses seed 42, whose validation difference (+0.00183) is the median of the three anchored-minus-baseline differences. The shown model pair is therefore illustrative only; the full evidence remains the seed-paired

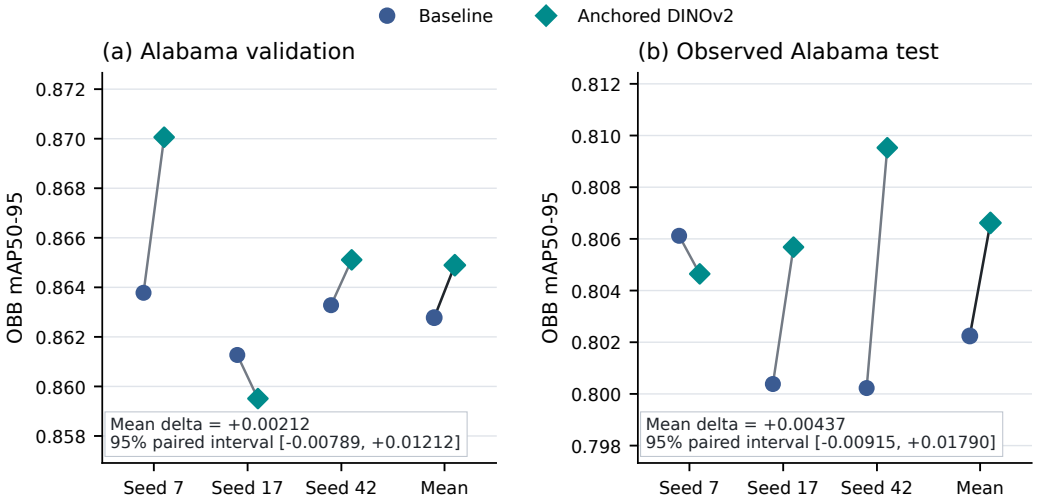


Figure 4: Paired three-seed comparison of the matched 1024-pixel baseline and domain-anchored DINOv2 initialization. Lines connect identical random seeds. Anchored SSL increased the mean on validation and on the observed Alabama test, but one validation seed declined and both paired 95% intervals include zero. The test panel is a held-file observation, not evidence of geographic generalization.

values in Table 2 and Figure 4. Both columns use the frozen validation-best checkpoints, a confidence threshold of 0.25, and rotated NMS at IoU 0.7.

5.4. Resolution, ensembling, and pseudo-labeling branches

The supporting branches refine, rather than change, that conclusion (Table 3). At 640 pixels, anchored SSL yielded a nearly neutral validation mean difference of +0.00061. The three-seed 640-pixel ensemble reached 0.86607 on validation, exceeding both the 640 baseline mean (0.86301) and the best individual 640 checkpoint (0.86479). This is a useful deployment candidate, but it has not been evaluated on an independent test and is not used as evidence that the main method generalizes.

The CTQ branch illustrates why a numerical validation gain is insufficient. In a separate 10%-labeled study, the selected CTQ pseudo-label configuration achieved 0.79646 on validation versus 0.78889 for its baseline. On the frozen observed test, however, CTQ reached 0.77668 versus 0.77942 for the baseline. The reversal is consistent with the retention audit: across 2,048 sampled Mississippi images per seed, only 4–7 images retained any pseudo box after confidence and non-maximum suppression. Retention was 0.195–0.342%, all

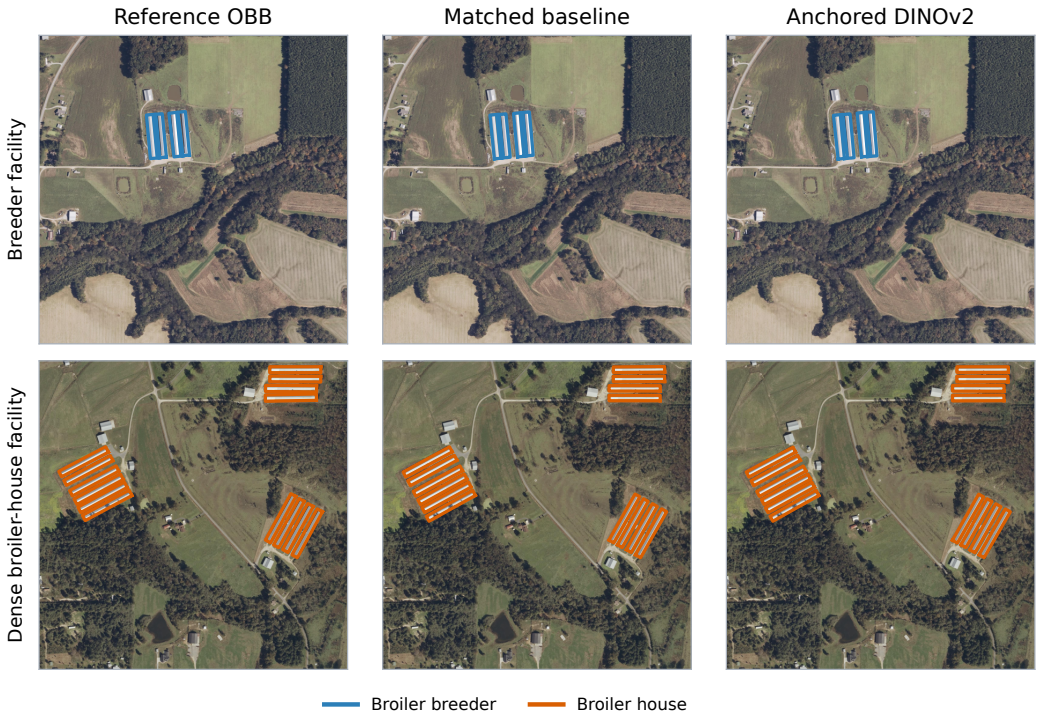


Figure 5: Qualitative oriented-detection behavior on two fixed Alabama validation images. Reference OBBs and predictions use the same class colors. The baseline and anchored models are the seed-42 frozen validation checkpoints, selected here by the median validation-effect rule; the scenes were selected by annotation content rather than model outcome. The panel is illustrative and does not replace the complete paired three-seed analysis.

retained boxes belonged to *broiler house*, and no *broiler breeder* boxes were retained. The bottleneck occurred before class-margin or two-view gates could improve quality; there was simply too little and too narrow a pseudo-label signal.

Table 3: Supporting experimental branches. Metrics are not pooled across rows because label fraction, image size, and evaluation phase differ. "Validation only" means no observed-test measurement was used or reported for that branch.

Branch	Development result	Observed-test status	Interpretation
1024 anchored DINOv2 SSL	0.86489 ± 0.00528 versus 0.86278 ± 0.00133 baseline	0.80662 ± 0.00257 versus 0.80224 ± 0.00336	Main methodological result; favorable, not conclusive
640 anchored DINOv2 SSL	0.86363 ± 0.00121 versus 0.86301 ± 0.00253 baseline	Validation only	Resolution follow-up; near-neutral effect
640 three-seed ensemble	0.86607; +0.00305 over baseline mean	Validation only	Deployment candidate; independent test still required
CTQ pseudo-labeling, 10% labels	0.79646 ± 0.01955 versus 0.78889 ± 0.00850 baseline	0.77668 versus 0.77942 baseline	Validation gain reversed; not selected
AL-to-MS pseudo retention	4–7 of 2,048 images retained labels per seed	No test involved	Pseudo-label signal too sparse and one-class

6. Discussion

6.1. What the anchored result means

The central outcome is modest in magnitude but methodologically informative. The proposed adaptation did not replace the detector with a DINO model, nor did it ask a detector to learn from high-confidence pseudo boxes in a shifted domain. It used a frozen DINOv2 teacher to supply dense aerial-image structure while the relative anchor restrained the backbone from moving arbitrarily far from its original detector state. This is a plausible mechanism for preserving useful OBB geometry while adapting the representation to regional aerial texture, field patterns, and roof context.

The observed pattern is compatible with that mechanism, but it does not prove it. The 1024-pixel three-seed means favored anchored SSL on both validation and the observed test. The slightly higher seed-to-seed variation

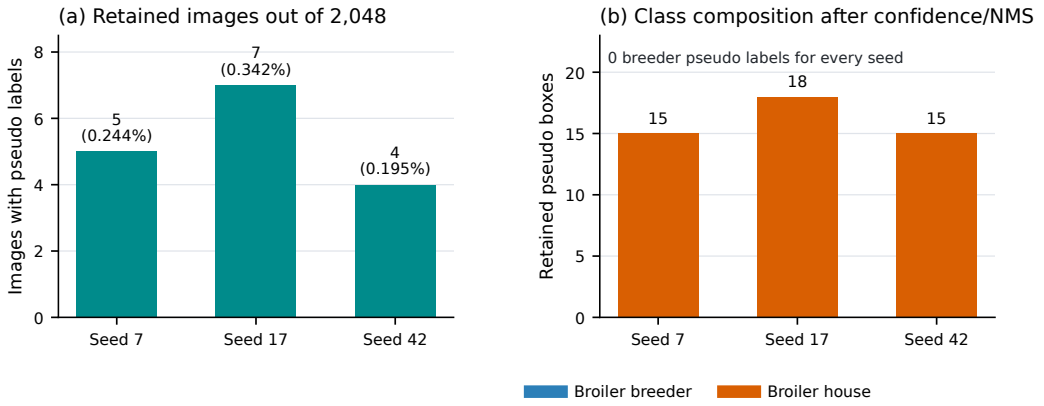


Figure 6: Pseudo-label retention audit on a fixed sample of 2,048 Mississippi images per seed. Only 4–7 images retained a pseudo-label after confidence and non-maximum-suppression filtering, corresponding to 0.195–0.342% image coverage. All retained boxes belonged to the broiler-house class. The sparse, class-collapsed signal explains why pseudo-label losses were not stacked with the proposed anchored representation method.

in the anchored arm, the negative validation delta for seed 17, and the wide paired intervals mean that the evidence is not yet sufficient to claim a reliable advantage. The rigorous conclusion is not "anchored SSL beats YOLO." It is that the method merits a confirmatory geographic evaluation because it showed a consistent mean direction without using test values for selection.

This distinction is valuable for agricultural AI. Much published model comparison is vulnerable to choosing a favorable seed, a favorable split, or a favorable post-processing threshold. Here, seed 42 did win on both observed phases, but it appears alongside all other seeds. The result remains credible precisely because the less favorable seed is visible and the protocol's own all-seed rule is reported.

6.2. Why pseudo-labeling was not stacked with the main method

It would be tempting to combine anchored SSL, EMA, CTQ, and pseudo-labeling into a single larger pipeline. The evidence argues against doing so. The pseudo-label audit was not merely lower-scoring; it diagnosed an absence of usable source-domain supervision. Less than four-tenths of one percent of sampled Mississippi images retained a box, and the retained labels contained only one of the two target classes. Adding a pseudo-label loss under those conditions would mostly add an unrepresentative signal and more

hyperparameters. It would also obscure the causal question addressed by the present method.

The negative CTQ result is therefore a useful result, not a failed appendix. It shows that a validation improvement can disappear when frozen checkpoints are evaluated later, and it identifies confidence and non-maximum suppression as the limiting stage. Future semi-supervised work should first establish adequate class-balanced retention, ideally with a small labeled Mississippi calibration set or an explicitly source-matched unlabeled corpus, before optimizing pseudo-label weights or consistency terms.

6.3. The ensemble is practical but secondary

Prediction-level ensembling of three 640-pixel baseline checkpoints increased validation mAP50-95 to 0.86607. For a production system, this may be an attractive accuracy-latency trade-off if hardware and throughput permit it. Scientifically, however, it is a secondary result. It is validation-only, differs in image size from the main 1024-pixel experiment, and does not establish that either a single model or an ensemble transfers beyond the current source structure. We treat it as a deployment candidate to be evaluated after the independent test is constructed, not as a substitute for the main contribution.

6.4. Limitations and confirmatory protocol

The most important limitation is test independence. The observed AL test is file-disjoint and was kept out of model selection, but its source-group overlap and perceptual similarity to development imagery make it unsuitable for a geographic generalization claim. This affects every method in the study, including the baseline. The next experiment should be planned before further tuning: lock the anchored SSL configuration, the matched baseline, and, if desired, the 640 ensemble; then evaluate them once on a test set split by farm, acquisition scene, or year with verified provenance and no shared source groups. The test set should contain both classes in sufficient number, and its imagery should be archived with a manifest that supports reproducible leakage auditing.

There are additional limits. The study uses three seeds, which is appropriate for an initial paired protocol but yields broad uncertainty intervals. The target task contains only two poultry-building classes, and the MS source has no labels that would quantify the actual domain gap. We also do not make claims about facility counting error, footprint area error, geographic completeness, or runtime at operational scale because those outcomes have

not yet been measured. Each should be reported in a future deployment study rather than inferred from mAP.

7. Conclusions

This study presents domain-anchored DINOv2 feature distillation as a focused representation-transfer method for oriented poultry-house detection. The method adapts a YOLO11l-OBB backbone on 250,000 unlabeled Mississippi aerial tiles while preserving its original detector structure through a relative parameter anchor. In a matched, three-seed Alabama experiment, it produced favorable mean mAP50-95 values on validation and on a frozen observed test. At the same time, the protocol's strict all-seed rule was not met, the paired intervals included zero, and the currently available test is not source-group independent.

The appropriate conclusion is constructive but restrained. Anchored DINOv2 adaptation is the strongest methodological direction in the present project because it has a transparent regional representation-learning rationale and survives comparison with matched controls more cleanly than pseudo-label alternatives. It is not yet ready to be described as a proven generalization improvement. A frozen, geographic or farm-disjoint confirmatory evaluation is the next decisive step. If the favorable direction remains under that test, the method will offer a strong and practical contribution to aerial agricultural infrastructure mapping.

Data and code availability

The analysis package contains resolved training configurations, SHA-256 manifests, protocol records, checkpoint identities, and aggregated metric files. The labeled and NAIP-derived imagery remain subject to their respective data-use terms. A public archival identifier should be assigned before submission.

Declaration of competing interest

The authors declare no competing interests.

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